BY ORDER OF THE COMMANDER
673D AIR BASE WING (PACAF)

673 AIR BASE WING INSTRUCTION
32-1007

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Civil Engineering

SAFEGUARDING UTILITIES FROM DAMAGE

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

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(Col Michael R. Staples)

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This instruction implements AFPD 32-10, *Installation and Facilities*, and establishes procedures to prevent damage to base utility systems, including communications, water, sewer, drainage, pavements, fire alarms, electricity, natural gas, environmental systems (systems and wells), and petroleum, oil, and lubricant (liquid fuel) systems caused by other than enemy action, ravages, and fair wear, and places responsibility for such damage. It applies to all utility systems within the geographical limits of Joint Base Elmendorf Richardson (JBER), and within areas where the 673d Air Base Wing (673 ABW) furnishes logistical support. This publication applies to US Army, US Air Force Reserve, and Air National Guard units and members. Submit requests for waivers through the chain of command to the publications OPR for non-tiered compliance items and this instruction cannot be supplemented or further extended. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the AF Form 847, *Recommendation for Change of Publication*. Route the AF Forms 847 through the appropriate chain of command. Ensure all records created as a result of processes prescribed in this publication are maintained in accordance with Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and disposed of in accordance with the Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS). The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

SUMMARY OF CHANGES

Reference to the Records publication, AFMAN33-363, in the purpose statement was rescinded. AFI33-322. *Records Management and Information Governance Program,* replaced the publication.

1. Procedures:

1.1. The offices listed below will maintain copies of appropriate maps and drawings, delineating the utility system under their control, and are authorized to approve work to be done on, and adjacent to, the controlled utility system:

1.1.1. **Commander, 673d Communications Squadron (673 CS).** Communication cable.

1.1.2. **Commander, 773d Civil Engineer Squadron (773 CES).** Steam, water, sewer, storm drainage, pavements, Runway Ice Detection System (RIDS) natural gas, fire alarms, cathodic protection, detection/alarm systems in airfield secure areas, liquid fuels, and electricity.

1.1.3. **Commander, 673d Civil Engineer Squadron (673 CES).** Environmental systems and clean-up sites.

1.1.4. **Anchorage Water & Wastewater Utility (AWWU).** Water and sewer lines under their control.

1.1.5. **Enstar Natural Gas ® (a trademark of Enstar Natural Gas Company).** Gas lines under their control.

1.1.6. **Anchorage Communications Systems (ACS) ® (a trademark of ACS).** Communication cables under their control.

1.1.7. **AT&T Alascom Communications ® (a trademark of Alascom Communications).** Communication cables under their control.

1.1.8. **General Communications, Inc. (GCI)** ® (a trademark of GCI). Cable TV systems under their control.

1.1.9. **Aurora Housing (Privatized Housing).** All utilities under their control.

1.1.10. **Doyon Utilities, LLC.** All utilities under their control.

1.2. All construction agencies, such as 673 CS, 773 CES, 673 CES, 176 CES, US Army Corps of Engineers, and all contractors will maintain close liaison with offices mentioned in **paragraph 1.1**., this instruction, to ensure no excavation is accomplished without prior coordination and approval for such excavation.

1.3. The construction agent will present a plan of intended work to the appropriate coordinating and approving base agency prior to commencing such work, to ensure existing base utilities will not be damaged. A duplicate copy of the construction drawing, if not already available to the base agency, will be made available by the construction agent. Any project in which mechanized equipment penetrates or disturbs the ground surface (including vacuum excavation), or any project in which hand digging will penetrate more than 4 inches into the ground, must be cleared in accordance with this instruction.

1.3.1. Formal clearances will be approved by the 773 CES Operations Flight commander or their designated representative(s) on 673 ABW Form 3, *Base Civil Engineer Work Clearance Request*. Clearance requests are not valid until signed by the approving officer and assigned a permit number. The approving officer will not sign the request until all utilities are marked, and all utility agency signatures are on the clearance request. Once completed and approved, the original work clearance request, with any applicable attachments, will be maintained on the job site by the requester during excavation and construction. An electronic copy of the approved work clearance request will be maintained in the 773 CES/CEOER office in accordance with appropriate records management table of disposition. When applicable, electronic copies will be uploaded to NexGenIt work task records to collect labor and material costs in accordance with NexGenIt business rules.

1.3.2. Scope of work will be identified on the clearance request. Drawings, sketches, and additional information will be attached to the clearance request. Any future changes to the scope of work will require a new clearance request.

1.3.3. Construction agencies requesting clearance will schedule a time to meet with each utility locate agency at the excavation site to locate affected utilities.

1.3.4. Construction agencies requesting clearance will be responsible for maintaining utility locates once they are initially marked by each utility agency by placing stakes in the ground or transcribing the marks on a drawing. If markings are damaged, construction agencies must request new locates and will give the utility locating agency at least 48 hours to perform utility locates. Expenses for time and materials to accomplish repeat locates will be borne by the construction agency at the respective shop rate of the locating utility owner and the actual cost of materials. Construction agencies will be provided drawings depicting the locations of environmental systems as an attachment to the original work clearance request. Systems and monitoring wells have been flagged, but construction agencies will be provided a 673 CES/CEI contact to call if additional information is required or flags are missing.

1.3.5. Excavation work will commence within 15 days after the approving officer's signature, otherwise the clearance request is invalid and a new request is required. The clearance becomes invalid 6 months after the approving officer's signature and must be renewed if the original requirement remains valid.

1.4. Agencies responsible for the movement of vehicles of unusual size or having cleat-type tracks will obtain a base map from the 773 CES Engineering Assistants (773 CES/CEOER) and coordinate a desired travel route with the agencies listed in **paragraph 1.1**, this instruction. When a travel route is approved, the agency will coordinate the vehicle movement with Security Forces personnel and provide an escort at the time of movement. Specific limitations will be in effect for:

1.4.1. Vehicles or loads in excess of 14 feet in height.

1.4.2. Any vehicles or objects which may damage the surface of pavements or grounds, due to the nature of track or skid.

2. Responsibilities:

2.1. All construction agencies will comply with this instruction.

2.2. All design and specification-preparing agencies will ensure all contractors are familiar with this instruction. This instruction must be referenced in all design specifications, and a copy will be provided to the contractor during the pre-construction meeting.

2.3. The responsible organization for each utility will:

2.3.1. Conduct investigations into the circumstances surrounding the damage of its facility, as to time, cause, responsible person, agency or activity, and determine whether proper prior clearance has been obtained.

2.3.2. Determine method of repair and cost, if the government is responsible, for time and material for repairs.

2.3.3. Initiate action against appropriate agencies for damage caused by individuals or agencies.

2.3.4. Initiate other action, as necessary, to minimize repetition or damage incident.

2.3.5. All utility agencies with utilities in the impacted area will locate their respective utility, regardless of the depth of excavation.

2.3.6. All utilities and excavation site will be marked according to the American Public Works Association uniform color codes:

2.3.6.1. **WHITE – Proposed Excavation (Black when snow is present).**

2.3.6.2. **PINK – Temporary Survey Markings.**

2.3.6.3. **RED – Electric Power Lines, Cables, Conduit, Lighting Cables.**

2.3.6.4. **YELLOW – Gas, Oil, Steam, Petroleum, or Gaseous Materials.**

2.3.6.5. **ORANGE – Communications, Alarms, or Signal Lines, Cables or Conduit.**

2.3.6.6. **BLUE – Potable Water.**

2.3.6.7. **PURPLE – Reclaimed Water, Irrigation, and Slurry Lines.**

2.3.6.8. **GREEN – Sanitary Sewers and Storm Drain Lines.**

2.3.7. When known, the total number of lines within the ground will be indicated. The number of lines indicated should be based on the physical lines. Multiple cables twisted together to form a single facility, as in the case of electric lines, would be considered one cable for locate purposes.

2.3.8. All agencies will notify the Fire Department, Hospital Emergency Room and Security Forces on all road closures.

2.3.9. To the extent possible, all agencies will allocate sufficient time to the GeoBase Section for the purpose of surveying/mapping of any exposed utility.

2.4. To minimize damage to the utility and loss of service to customers; all digging within 2 feet of the marked utility will be performed by hand or vacuum excavation, according to industry practices.

2.4.1. Some utilities, that is, certain communications cables or electrical lines, may require hand digging within 3 or 4 feet. This requirement will be at the discretion of the Commander, 673 CS, 673 CES, 773 CES, or each utility owner as appropriate.

2.4.2. Under certain conditions, digging could be accomplished within 2 feet of a utility with equipment, upon written request and approval. This requirement will be at the discretion of the Commander, 673 CS, 673 CES, 773 CES, or each utility owner as appropriate.

2.4.3. ENSTAR ® has specific safety requirements for excavations adjacent to natural gas pipelines.

2.5. Any damage to utilities will be the responsibility of the construction agency. Repairs to damaged utilities will commence within 2 hours, unless otherwise specified by the utility owner. Communication cable repairs will be performed by a licensed and certified technician, unless otherwise specified by the utility owner.

2.6. Coordination of oversized vehicles and equipment movement will be through either the 673 CES Construction Management section or the 773 CES section requiring vehicle or equipment movement. The responsible agency will notify the 673d Security Forces Squadron Law Enforcement Desk prior to such a movement, and will ensure the contractor or agency moving the oversized vehicle or equipment provides an escort and spotters to prevent damage to exposed utilities such as overhead cables, fire hydrants, utility poles, and so forth.

2.7. The Chief, Fire and Emergency Services Flight will coordinate on all construction activity involving disruption of utility services--especially water supply, fire alarms/suppression systems, and street closures. NOTE: Disruption of water supply will be delayed during extremely dry climatic conditions, near open fields, tree areas, and so forth, when requested by the Fire and Emergency Services Flight.

2.8. All self-help involving excavation, installation of television/citizen's band radio antennas, or any other work that may involve disruption of utilities requires the requester to have an approved 673 ABW Form 3 which can be obtained from the 773 CES Customer Service Unit and coordinated through appropriate agencies.

2.9. The Air Force, the Environmental Protection Agency (EPA), and the Alaska Department of Environmental Conservation (ADEC) have signed Records of Decision (ROD) that outline land use control (LUC) requirements for contaminated sites on JBER. All construction agencies will comply with the general land use controls as described below (Note: There are specific land use controls associated with certain environmental sites, which will be provided to the construction agency at the time of Work Clearance Request review):

2.9.1. Stop work and immediately contact 911 if contaminated soil or debris is encountered during excavation. Contaminated soil, if encountered, must be handled in accordance with the current JBER Hazardous Waste Management Plan. The construction agency is responsible for proper handling until laboratory analysis determines otherwise.

2.9.2. Immediately contact Environmental Restoration, 384-2984, if an environmental system is encountered or damaged.

2.9.3. Prepare a Work Plan, with Health and Safety Plan and Sampling and Analysis Plan, if working in an area of known contamination. Obtain EPA and ADEC approval prior to beginning work.

2.9.4. Water from the shallow aquifer is not allowed to be used for any purpose including, but not limited to, drinking, irrigation, fire control, or dust control.

2.9.5. Submit for approval to Environmental Restoration a report of engineering controls and methods designed to prevent cross contamination from the shallow aquifer to the confined aquifer prior to submitting a work clearance request to drill through the shallow aquifer.

2.9.6. Do not damage or interfere with access to and operation of any environmental system.

2.9.7. If ground disturbance will occur in urbanized area, the excavator must submit a Storm Water Pollution Prevention Plan (SWPPP) for projects greater than or equal to 1.0 acre or an Erosion and Sediment Control Plan (ESCP) for projects less than 1 acre to JBER Water Quality Program Manager for review (**673CES.CEIEC.EnvCom@us.af.mil**) at least 14 calendar days prior to commencing activity.

2.9.8. Return to Environmental Restoration a written certification of compliance with these land use controls no later than 30 days upon completion of any excavation or drilling.

2.9.9. Stop work immediately and contact CES/CEIEC 384-3467 if bones, artifacts, bottles, cans, stone tools, or structural remains are encountered during excavation.

3. Information Collections. No information collections are required by this publication.

PATRICIA A. CSÀNK, Colonel, USAF
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI33-322, *Records Management and Information Governance Program, 23 March 2020*

AFPD 32-10, *Installation and Facilities*, 3 March 2010

JBER OPLAN 19-3, *Environmental Management Plan*, 1 June 2016

Adopted Forms

AF Form 847, *Recommendation for Change of Publication*

Prescribed Forms

673 ABW Form 3, *Base Civil Engineer Work Clearance*

Abbreviations and Acronyms

ABW—Air Base Wing

ACS—- Anchorage Communications Systems

ADEC—- Alaska Department of Environmental Conservation

AFI—- Air Force Instruction

AFRIMS—- Air Force Records Information Management System

CES—- Civil Engineer Squadron

CS—- Communications Squadron

EPA—- Environmental Protection Agency

GCI—- General Communications

LLC—- Limited Liability Corporation

LUC—- Land Use Control

OPLAN—- Operations Plan

OPR—- Office of Primary Responsibility

RDS—- Records Disposition Schedule

ROD—- Records of Decision